High-Dimensional Feature Spaces
Redundancy, Collinearity, and Reduction

Feature Selection – Core "make-or-break" Steps in both Econometrics and ML

Guiding principle

- **Match feature selection to the modelling objective**
- **Inference:** estimate *effects* (sign, magnitude, confidence)
- **Prediction:** maximize out-of-sample performance (often under distribution shift)

Goal

- **Reduce redundant** & **stable feature set** that:
- **Maximizes independent information content** – reduce redundancy
- **Robust across time/regimes**

Key concerns in high-dimensional financial feature spaces

- **Redundancy / strong correlation -** many features carry the same signal (lags/derivatives/technicals)
- Adding complexity **without adding information**
- Increased variance of estimates and fragility of feature rankings

Multicollinearity (especially for linear/econometric models)

- Coefficients become **unstable** (sign/magnitude etc.)
- Many models fit almost equally well
- Can show **high** R^2 yet **insignificant individual coefficients**

Feature Selection – Core “make-or-break” Steps in both Econometrics and ML

Different target types require different thinking

1. **Continuous target (e.g., forward return)**
   - **Objective:** predict **magnitude** (and possibly sign)
   - **Emphasis:** stability of expected value

2. **State target (e.g., bull/bear classification)**
   - **Objective:** predict **probability of regime / direction**
   - Emphasis: separability, decision thresholds, class imbalance, temporal dependence
   - **Features that separate regimes may differ from those that predict continuous returns**

Exercise

Original Input Data – Rank Percentiles

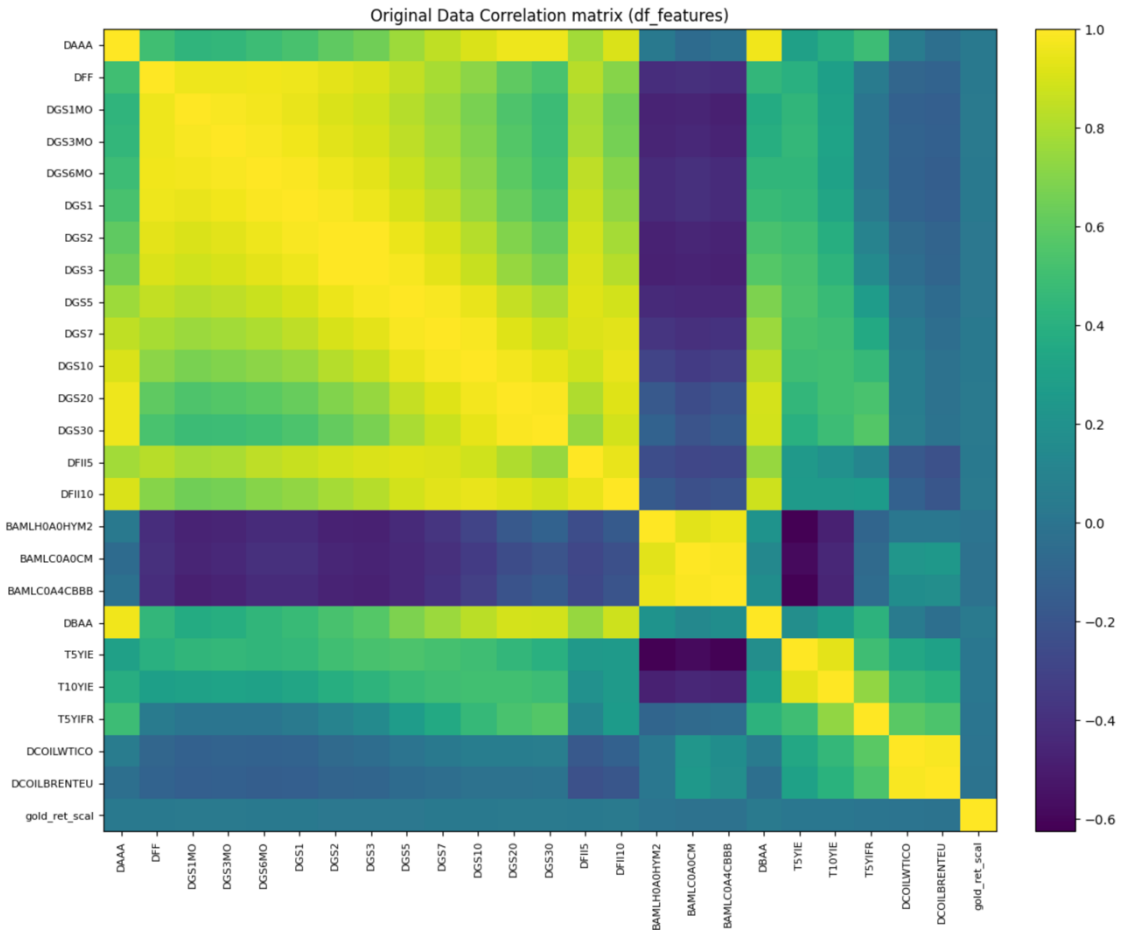
Feature	Category	Business description
DCOILBRENTEU	Commodities — Energy	Brent crude oil spot price (Europe).
DCOILWTICO	Commodities — Energy	WTI crude oil spot price (Cushing, OK).
gold_ret_scal	Commodities — Precious metals (derived)	Scaled gold returns (your engineered/standardized gold return series).
DAAA	Credit — Corporate yields	Moody's Seasoned Aaa Corporate Bond Yield (high-grade corporate yield level).
DBAA	Credit — Corporate yields	Moody's Seasoned Baa Corporate Bond Yield (lower IG; more cyclical).
BAMLC0A4CBBB	Credit — OAS (BBB)	ICE BofA BBB US Corporate Index Option-Adjusted Spread (BBB subset of IG).
BAMLH0A0HYM2	Credit — OAS (HY)	ICE BofA US High Yield Index Option-Adjusted Spread (HY/junk spread).
BAMLC0A0CM	Credit — OAS (IG)	ICE BofA US Corporate Index Option-Adjusted Spread (investment-grade spread).
T10YIE	Inflation expectations	10-Year breakeven inflation rate (nominal 10Y vs 10Y TIPS implied inflation).
T5YIE	Inflation expectations	5-Year breakeven inflation rate (nominal 5Y vs 5Y TIPS implied inflation).
T5YIFR	Inflation expectations	5-year, 5-year forward inflation expectation (expected inflation years 5–10).
DFF	Rates — Policy / Money market	Effective Federal Funds Rate (overnight policy implementation rate).
DFII10	Rates — Real (TIPS)	10-Year TIPS constant maturity real yield.
DFII5	Rates — Real (TIPS)	5-Year TIPS (inflation-indexed) constant maturity real yield.
DGS1	Rates — UST nominal curve	1-Year Treasury constant maturity yield.
DGS10	Rates — UST nominal curve	10-Year Treasury constant maturity yield (benchmark long rate).
DGS1MO	Rates — UST nominal curve	1-Month Treasury constant maturity yield.
DGS2	Rates — UST nominal curve	2-Year Treasury constant maturity yield.
DGS20	Rates — UST nominal curve	20-Year Treasury constant maturity yield.
DGS3	Rates — UST nominal curve	3-Year Treasury constant maturity yield.
DGS30	Rates — UST nominal curve	30-Year Treasury constant maturity yield.
DGS3MO	Rates — UST nominal curve	3-Month Treasury constant maturity yield.
DGS5	Rates — UST nominal curve	5-Year Treasury constant maturity yield.
DGS6MO	Rates — UST nominal curve	6-Month Treasury constant maturity yield.
DGS7	Rates — UST nominal curve	7-Year Treasury constant maturity yield.

Feature Shape : (5605, 26)

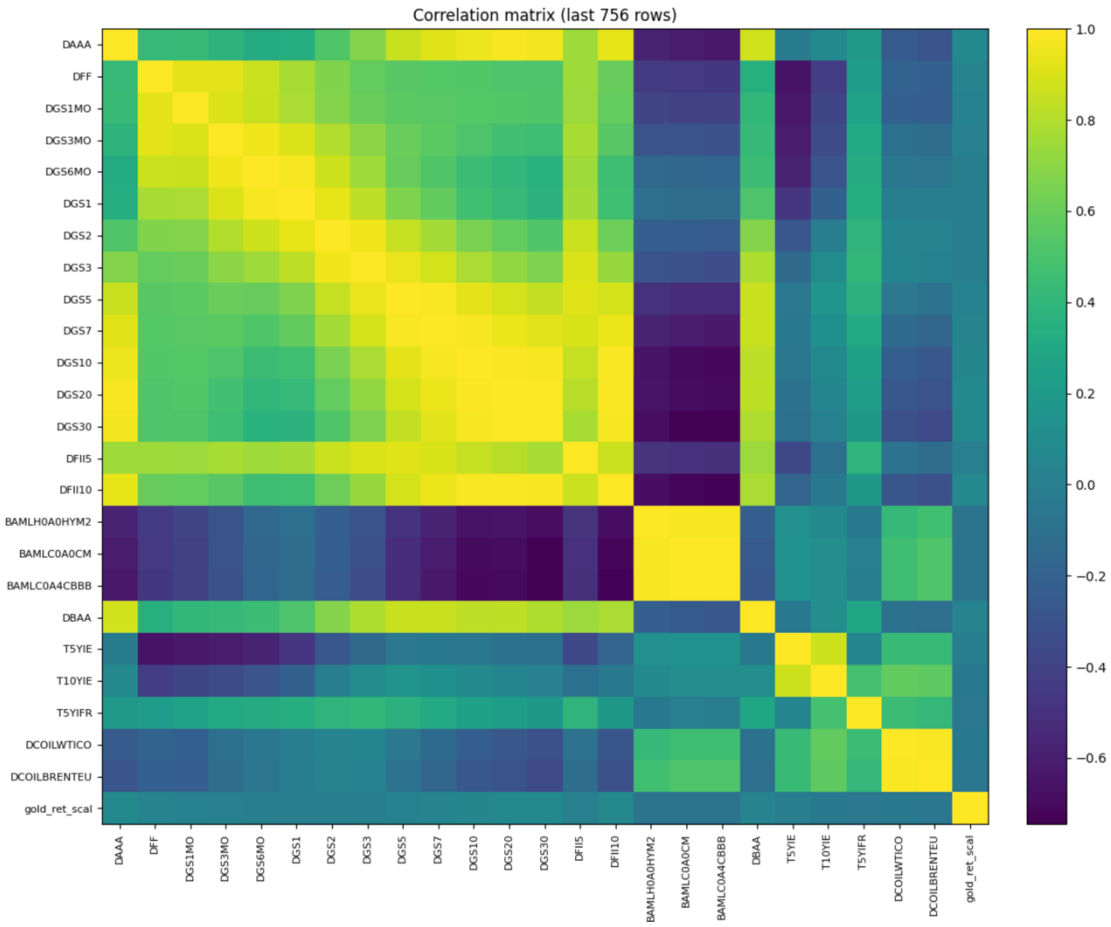
Target Shape : (5617, 1)

Exercise

Original Input Data (spearman Correlation)



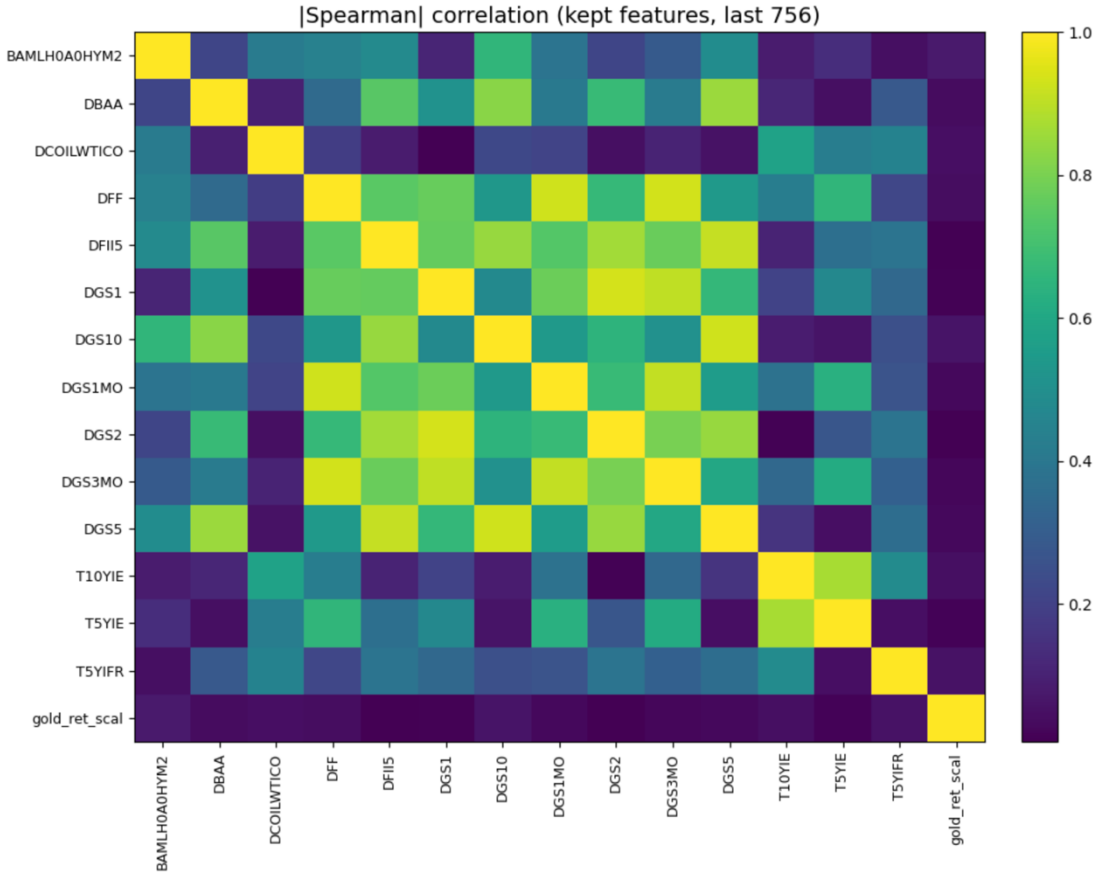
Subset Last 3 years , x756 observations



Multicollinearity – Method 1

Pairwise similarity + clustering approach to **redundancy elimination i.e. unsupervised redundancy removal**

Kept Features Only



- **Rank Percentiles**, **spearman correlation**: measures how similarly two features move in rank space (monotonic association)
- **Convert correlation redundancy into a distance d=1-| |, so (1-0.95)**
- **Hierarchical clustering** builds dendrogram by merging the closest clusters
- **fcluster(..., t=0.05, criterion="distance") cuts the dendrogram &** yields redundancy clusters
- **Keep one "Medoid" representative** per redundancy cluster
- print(corr_keep.shape) = (15, 15)

Method 1 - Summary - Pairwise Similarity + Clustering redundancy elimination

- **Rank percentiles** Input Data: Numeric scale is **ordinal**: 0.92 means "higher than 92% of history"
- **Association measure: - Spearman Correlation is the natural dependence measure**
- **Approach Model-free**: doesn't assume any downstream modelling
- **Interpretation**: Identifies "redundant groups" that can be justified (WTI vs Brent etc.)
- **Quickly identifies near-duplicate features (highly correlated series)**
- **Scales nicely**: for 25–300 features it's fast and easy to visualize
- **Aligns with business intuition**: Clusters often map cleanly to specific blocks

Some Limitations

- **It is fundamentally pairwise:** Pairwise correlation can miss multi-way redundancy
- **Weakly correlations** possible with any other feature
- **True multicollinearity can be missed**
- **Clustering depends on threshold and window :** corr_thresh=0.95 vs 0.90, "last 756" - matters of judgement

Notes:

- **Correlation-based clustering doesn't "solve" collinearity in the strict algebraic sense**
- **Selected feature sets for return regression and bull/bear classification can overlap, but should not be assumed identical**

Variance Inflation Factor

- VIF - **diagnosing** multicollinearity for **linear models**
- VIF Measures:
  - VIF = 1 - no multicollinearity
  - VIF 1 - 5 - acceptable multicollinearity
  - VIF > 5 - concerning multicollinearity
  - VIF > 10 - severe multicollinearity

Many methods of VIF computation:

- Relaxed screening
- Progressive selection
- **Adaptive Thresholds**
- Cluster based selection

Step 1: Using core base data only

Feature Shape : (5605, 26)

Output:

	Feature	VIF Score
1	gold_ret_scal	1.00
2	BAMLH0A0HYM2	5.05
3	DFF	5.2
4	DCOILWTICO	93.5
5	DCOILBRENTEU	94.7

Variance Inflation Factor

Step 2: Adding Derived Features

- _raw – all base features
- _diff22 - df_raw[col].diff(22)
- _z252 - z_score(252)

- diff22 - breaks "level" collinearity
- z252 - mixes in volatility/regime information

Expanded Data Set New Input

Feature	VIF
gold_ret_scal_diff22	2.3
DFF_diff22	8.5
DGS1MO_diff22	8.51
DFF_z252	10.58
BAMLH0A0HYM2_diff22	21.06
DGS1MO_z252	32.79
DGS3MO_diff22	38.55
DCOILBRENTEU_diff22	46.98
DCOILWTICO_diff22	47.31
DGS6MO_diff22	97.03
DGS1_diff22	106.99
T5YIFR_z252	133.34
T5YIE_z252	147.69
gold_ret_scal_z252	166.43
gold_ret_scal_raw	169.63
DAAA_diff22	194.44
T10YIE_z252	198.32
DGS3MO_z252	212.68
BAMLH0A0HYM2_z252	226.06
DCOILBRENTEU_z252	253.58
DGS1MO_raw	255.76
DGS2_diff22	308.17
DCOILWTICO_z252	345.1
DFF_raw	384.37
BAMLH0A0HYM2_raw	433.22
BAMLC0A0CM_diff22	476.72
BAMLC0A4CBBB_diff22	506.16
DBAA_diff22	506.48
DBAA_z252	532.77
DGS3_diff22	549.69
DGS20_diff22	603.13
DGS6MO_z252	604.16
DCOILBRENTEU_raw	652.08
DGS30_diff22	763.13
DCOILWTICO_raw	781.97
DFII5_z252	841.69
DAAA_z252	1159.14
DFII10_z252	1291.47
DBAA_raw	1451.8

Generated from 25 -> 75 features

Final VIF Selection VIF < 50

Rank	Feature	VIF	Quality
1	gold_ret_scal_diff22	1.03	Excellent
2	BAMLH0A0HYM2_diff22	1.22	Excellent
3	T5YIE_z252	1.48	Excellent
4	T5YIFR_z252	1.54	Excellent
5	DGS1MO_diff22	2.27	Excellent
6	DFF_diff22	2.86	Excellent
7	DFF_z252	6.22	Excellent
8	DGS1MO_z252	7.29	Excellent
9	DGS1_diff22	9	Excellent
10	DGS3MO_diff22	9.06	Excellent
11	DCOILBRENTEU_diff22	10.91	Good
12	DCOILWTICO_diff22	12.38	Good
13	DGS6MO_diff22	22.23	Good

Base variables that are structurally linked:

- Treasury curve points (DGS1MO, DGS3MO, DGS1, DGS2, …)
- Inflation expectations (T5YIE, T10YIE, T5YIFR) are also tightly linked

Base space, measures the same latent drivers multiple times

Two series that share the same level can still have different:

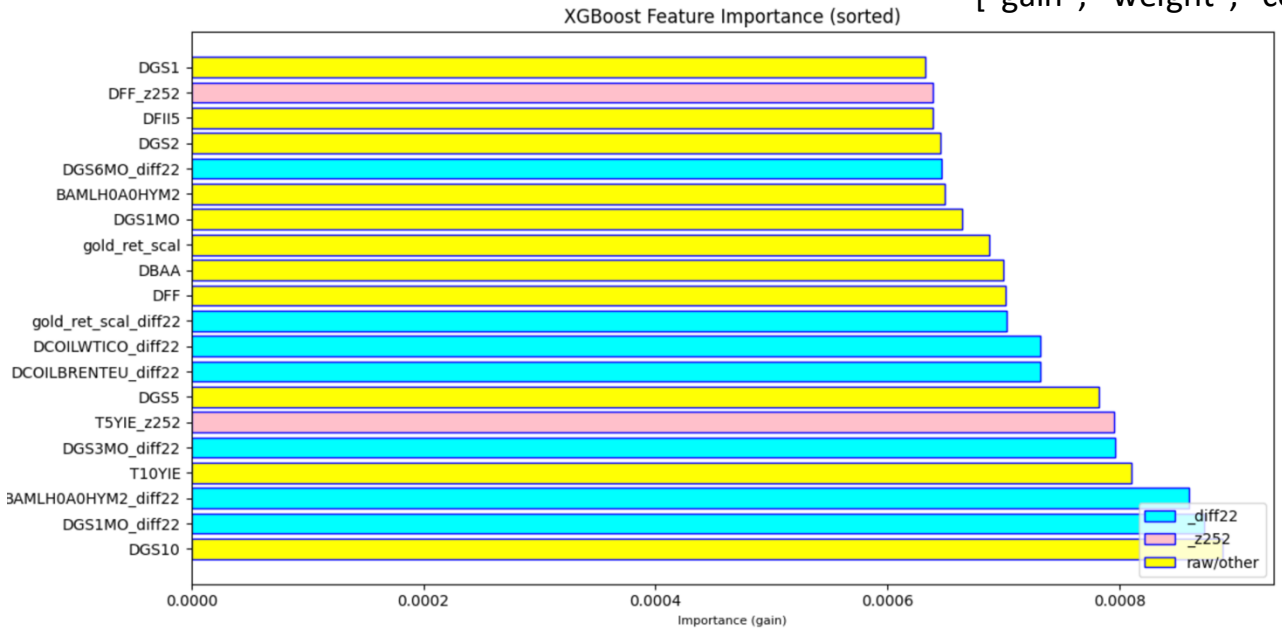
- recent drift relative to their own history
- volatility scaling
- mean reversion timing

XGBOOST - Combining Pairwise similarity + clustering +VIF

Workflow:

- **Step – 1: Pairwise redundancy (Spearman + clustering)** → drop near-duplicates (keep 1 rep per cluster)
- **Step – 2: VIF on the resulting candidate set** → drop multi-way collinearity (keep VIF < 50)
- **Step - 3: Concatenate Step1 & Step2 data** in X and assign target variable y (continuous)
- **Step – 4: Fit XGBoost** on the final X (+ y) - get feature importance

["gain", ~~"weight", "cover"~~]



XGBoost assigns importance to **whichever representation of a variable best improves prediction at the margin**:

- **basse series (levels)** carry slow-moving state information (e.g., where rates/spreads/oil currently are)
- **diff22** features carry **recent shocks and momentum** (how fast the state is changing over ~1 month)
- **z252** features express **relative extremeness** (how unusual today is versus a long historical baseline)

High-Dimensional Feature Spaces: Redundancy, Collinearity, and Reduction

Dimensionality reduction and multicollinearity control in high-dimensional feature spaces is rarely a choice of *one* method; it is typically a **pipeline** and an **iterative process**

Some techniques are primarily **diagnostic** (they reveal redundancy and instability), while others do the **heavy lifting** (they enforce sparsity, compress information, or regularize the model)

The right sequence depends on the end goal—**inference vs prediction, interpretability vs accuracy**, and robustness under regime shifts.

In practice you combine tools such as:

- **pairwise redundancy checks and clustering**
- **VIF** (multi-way collinearity)
- **regularization** (Elastic Net)
- **model-based selection** (e.g., XGBoost importance/SHAP)
- **SVD/PCA/PLS -** true compression methods
- **mRMR -** constraint-based method

Walk-forward evaluation: run diagnostics/selection **within each rolling/expanding training window** (and confirm stability over folds/regimes) to avoid look-ahead bias and brittle feature sets

Dimensionality reduction is a core modelling function to **reduce complexity** without **discarding genuine signal, validate** all choices